

Level		OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last Updated	Source(s)	System/ Subsystem	Status?	VERIFICATION & VALIDATION	
OBJ = Objective, REQ = Requirement, CSR = Constraint. Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)			Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measureable, and testable guideline to be met.	Date of when the requirement statement was last revised.	Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?	Accomplished or unaccomplished?		
MISSION STATEMENT			The multi-agent Europa surface system shall autonomously characterize the ice shell and identify a cryobot insertion site, operating without real-time Earth support.	6/14/2026	-	-	-		
OBJ.01	System Level		The system shall demonstrate autonomous multi-agent coordination and consensus-based decision making without Earth communication links	6/14/2026	This objective addresses the multi-agent coordination shortfall ranked 139th in NASA's Civil Space Technology Shortfall Rankings, which identifies cooperative task planning as a near-term priority (STMD) This objective addresses NASA's identified gap in autonomous navigation without Earth-based support, as stated in the FY2026-2027 SBIR solicitation Topic COMNAV.1.S26B [SBIR] This objective supports the science goals outlined in the Decadal Survey, which identifies Europa's subsurface ocean as a primary target for astrobiology and calls for characterization of the ice shell to assess habitability. [OWL] This objective addresses NASA's identified gap in autonomous distributed spacecraft operations without Earth-based navigational support, per SBIR Topic COMNAV.1.S26B. [SBIR]				
OBJ.02	Subsystem Mobility		The rover shall traverse Europa's surface terrain and maintain localization using onboard sensors and inter-agent ranging	6/14/2026					
OBJ.03	Subsystem Science/Payload		The system shall characterize ice shell thickness using a ground penetrating radar sounder and identify candidate insertion sites	6/14/2026					
OBJ.04	Subsystem COMMs		The inter-agent messh network shall maintain coordination under degraded connectivity and radiation induced faults	6/14/2026					

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Verification - Does it meet the specification need? Are we building our product right?

Validation - Does it meet the customer need? Did we build the right product?

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